

Module 5 — Patent Landscape Reports

45-minute block · TIP4PATLIBS course material

Landscape analyses after **Riccardo Priore**, Centro PATLIB, AREA Science Park. This module is contributed material, reworked to match the course's look. The analytical approach and the worked example are his.

How to read this document. The running text addresses **you, the participant**. Boxes labelled **Trainer** are for whoever is **running the session**; boxes labelled **Trap** are traps that have caught people before.

Learning objective

I can define a patent corpus with a search strategy I am able to defend, run a battery of standard analyses over it, and assemble the result into one self-contained report a client can keep.

Prerequisites

- **Module 1** — TIP is running, PATSTAT connects.
- **Module 2** — you can read a query and judge a result.
- **Module 3** — you know that an applicant name is not a company.

Sub-objectives

By the end you can:

1. **Build a corpus as an intersection**, not as a keyword search: keywords **AND** (IPC OR CPC), and explain what each half of that rule is for.
2. **Justify an exclusion** — say why generic terms and bare acronyms were deliberately left out, and what precision that bought.
3. **Recognise the report pipeline's one shape**: a question in markdown → one code cell that reads the shared corpus, queries PATSTAT and builds one figure → one `record(...)` call.
4. **Read a landscape report critically** — which of its thirteen charts answers *who*, which answers *where*, which answers *when*, and which answers *how it connects*.
5. **Say what the assembled artifact is**: one self-contained HTML file plus one workbook, no internet required.

Material

Folder	5_patentreports/2_antibiotic_resistance_rebuild/
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Notebooks	1_dataset_and_search_strategy · 2_core_landscape_analyses · 3_advanced_analyses · 4_assemble_report
Artifact	4_report/antibiotic_resistance_report.html + ..._report_data.xlsx
Runs on	EPO TIP, PATSTAT PROD (notebook 2's citation analysis uses BigQuery)
Ships	pre-executed — 29 of 29 code cells, outputs kept

Trap — This module ships with its outputs, and that is deliberate. Modules 1–4 clear their outputs because participants are meant to run them. Module 5 is read as a **finished report** in a showcase. The outputs *are* the deliverable — never re-run the cells to "tidy" them.

Trainer. The sibling folders 1_antibiotic_resistance/ (the imported reference) and 2_antibiotic_resistance_mvp/ (the frozen MVP) are **not** course material. Use the rebuild. If someone asks about t-SNE clusters or triadic families, those live in the reference folder and are a future phase of the rebuild — say so rather than improvising.

Phase 1 · Introduction (≈ 7 min)

The question this module answers

A regional cluster manager asks: "give us a picture of what is happening in antibiotic resistance." Where do you start?

Everyone starts in the same place — a keyword — and this is where a landscape report is won or lost. Ask the room what they would search for. Somebody will say **"drug resistance"**. Somebody will say **MRSA**.

Both are the wrong answer, and for opposite reasons:

- **"drug resistance"** is dominated by **cancer**. Your antibiotic landscape would be an oncology landscape wearing the wrong label.
- **MRSA, VRE, ESBL** — bare acronyms — are noisy. They collide with unrelated strings and appear in documents that are not about the thing at all.

Neither problem is visible in the result. You get a large corpus, it looks healthy, and it is about something else.

The fix is not a better keyword. It is a **rule with two independent halves**: a family must match a keyword **and** carry a relevant classification. Keywords bring recall; classification brings precision. Neither alone is defensible.

Teaching and learning activity		Time
Opening	Trainer asks for search terms for "antibiotic resistance"; collects them	2 min
Tension	Trainer names what "drug resistance" and bare acronyms actually return	3 min
Framing	Trainer states the module's rule: keywords AND (IPC OR CPC) — and that the exclusions are part of the method, to be written down	2 min

Phase 2 · Working through (≈ 28 min)

Four notebooks, one pipeline. **Run them in order 1 → 2 → 3 → 4** — each writes what the next reads.

Step 1 — Define the corpus (10 min) · `1_dataset_and_search_strategy.ipynb`

Eight steps that end in one file.

What happens	
1	Connect to PATSTAT PROD via the SQLAlchemy ORM interface, and import <code>report_kit</code>
2	The keyword strategy — three groups of terms: the core concept, the resistance <i>mechanisms</i> , and an explicit bacterial context. Generic "drug resistance" and bare acronyms are excluded on purpose
3	Classification filters (IPC / CPC) , one code list serving both
4	Combine : intersect the keyword families with the classification families
5	One representative publication per family , 2000 onwards, preferring EP > WO > US
6	Export <code>dataset.xlsx</code> — the single shared corpus notebooks 2 and 3 read
7	First chart — how the field has grown, by earliest filing year
8	Second chart — the twenty leading IPC classes

Trap — Watch the spacing in IPC symbols. An 8-character IPC symbol is a 4-character sub-class, then padding spaces, then the main-group number: A61K 31 is **two** spaces, C12Q 1 is **three**. Get the padding wrong and the filter matches nothing — silently, with no error.

Trainer. Step 5 is a decision worth naming: *one representative publication per family*, preferring EP then WO then US. That is a choice, and a different preference order would give a different citable number for the same invention. It is the module 3 lesson again — the defensible part is that it was written down.

Step 2 — The core landscape battery (10 min) · `2_core_landscape_analyses.ipynb`

Ten charts, all reading the same corpus. Do not walk through all ten — walk through the **structure**, then let the room read.

The question	The charts
Where?	filings by international/regional authority · national filing trends · national vs international filing strategy
When?	WO (PCT) vs EP over time · innovation waves by technology area
How far?	family size & global reach
Who?	top applicants by families · applicants by institutional sector · grant rate by top applicant · most influential organisations (forward citations)

Trainer. The sector chart and the grant-rate chart are the two a PATLIB audience reacts to, because they separate universities from companies and separate *filing a lot* from *getting granted*. If time is short, spend it on those two and let the rest be read.

Trap. The forward-citation analysis runs against **BigQuery**, not the TIP PATSTAT client — a citation self-join is the honest tool for that question and BigQuery is where it is affordable. That cell therefore needs separate credentials and will not run for a participant who only has TIP. Its output ships with the notebook.

Step 3 — One advanced analysis (4 min) · `3_advanced_analyses.ipynb`

A **technology co-occurrence network**: which IPC fields appear together on the same families. Five steps — load the corpus, map IPC codes to technology fields, keep the codes with at least 50 families, self-join to find co-occurrences, draw the network above an edge threshold.

This is the one chart in the report that answers a question the others cannot: **how the field is connected**, rather than how big it is or who is in it.

Step 4 — Assemble the report (4 min) · `4_assemble_report.ipynb`

Four steps, and this is where the pipeline pays off.

Every analysis in notebooks 1–3 ended with a `report_kit.record(...)` call that showed the figure inline *and* saved two things: the figure as an inline HTML fragment, and the data behind it. Notebook 4 collects every contribution from the output manifests, orders them, and stitches them into:

- **one self-contained HTML report** — a single embedded copy of `plotly.js`, **no iframes**, so it renders inside TIP and works with no internet at all. It has two view modes switched by a header button: **Paged** (one chart at a time, step bar, Previous/Next, ←/→ keys) and **One page** (everything stacked).
- **one data workbook** — one sheet per chart, so a client can check any number.

It is opened with the course's shared `open_html()` helper.

Trap — Never open a TIP-generated report with `IFrame`. TIP's content-security policy sandboxes iframes and disables their JavaScript, so an interactive report renders as a blank or dead box. `open_html()` goes through jupyter-server-proxy and works. This applies to modules 6, 7 and 8 alike.

Step	What you do	What you see	Time
1	Build the corpus	One <code>dataset.xlsx</code> , and two charts	10 min
2	Run the core battery	Ten charts: where, when, how far, who	10 min
3	Run the network analysis	How the technology fields connect	4 min
4	Assemble	One HTML report + one workbook	4 min

Phase 3 · Learning outcome (≈ 10 min)

What now exists

- **One self-contained HTML landscape report — 13 charts** (2 from notebook 1, 10 from notebook 2, 1 from notebook 3) — that opens on any machine with a browser and no internet.
- **One workbook** carrying the data behind every chart.
- **A written search strategy** — including its exclusions — that you can hand to anyone who disputes the corpus.

The third item is what makes the first two worth anything.

Self-check

1. **Why is `keywords AND (IPC OR CPC)` better than either half alone?** (*Keywords give recall but no precision; classification gives precision but misses anything classified oddly. The intersection is defensible in both directions.*)
2. **You are asked why `MRSA` is not in your search terms.** Your answer? (*Bare acronyms are noisy; precision comes from requiring a classification match. Note that this is an argued choice, not an oversight.*)
3. **Your IPC filter returns zero families and throws no error.** (*Check the padding: `A61K 31` is two spaces, `C12Q 1` is three.*)
4. **A client opens your report on a laptop with no internet. Does it work?** (*Yes — the plotly library is embedded and there are no external requests. That was a design requirement, not a convenience.*)

Transfer to your own work

Take the pipeline and change **only notebook 1**. Pick a technology your region actually cares about, write the three keyword groups and the classification list, and run notebooks 2–4 unchanged.

That is the real lesson of the module: the analyses are generic; **the corpus definition is the part that is yours**, and it is the part you have to be able to defend.

Trainer. Close by opening the shipped report in Paged mode and walking the last three slides of it. The participants have just seen how each chart is made; seeing them assembled as a client-facing document is what makes the pipeline feel worth building.

Where this leads

Next	Why
Module 6 — IPScore	Module 5 describes a whole field. Module 6 asks what one patent in it is worth — and how much of that answer is evidence. The worked example in module 6 is drawn from <i>this</i> corpus.
The IPScore reference — IPScore (Riccardo's tools)	The companion valuation module, outside this course block.

Notes for the next revision

- The rebuild's `README.md` states that notebook 3 holds the technology network only, and that temporal, citation, t-SNE, SDG and triadic analyses are "the next phase". The repository-level `CLAUDE.md` describes module 5 as including t-SNE clusters and triadic families — that describes the older `1_antibiotic_resistance/` folder. **Align the two descriptions** before the workshop so nobody promises a chart that is not in the report.
- The forward-citation cell's BigQuery dependency should be stated in the notebook header, not only in the cell comment — a participant who tries to re-run notebook 2 on TIP alone will hit it.